

Examples of working with variance and covariance

① Suppose $X_1, \dots, X_n$ are iid with mean μ and variance σ^2
 $E[X_i] = \mu$ $\text{var}(X_i) = \sigma^2$ $\forall i \in \{1, \dots, n\}$

(a) Consider the sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

We found that $E[\bar{X}] = \mu$ and $\text{var}(\bar{X}) = \frac{\sigma^2}{n}$ ✓

(b) Consider the sample variance $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$

~~$(X_1 - \bar{X})^2 + (X_2 - \bar{X})^2$~~

$$E[S^2] = E\left[\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2\right] \quad E[(X_i - \bar{X})^2]$$

$$= \frac{1}{n-1} \sum_{i=1}^n E[(X_i - \bar{X})^2]$$

$$= \frac{1}{n-1} \sum_{i=1}^n E[(\underbrace{X_i - \mu}_a + \underbrace{\mu - \bar{X}}_b)^2]$$

$$= \frac{1}{n-1} \sum_{i=1}^n E[\underbrace{(X_i - \mu)^2}_{a^2} + 2 \underbrace{(X_i - \mu)}_a \underbrace{(\mu - \bar{X})}_{b^2} + (\mu - \bar{X})^2]$$

$$= \underbrace{\frac{1}{n-1} \sum_{i=1}^n E[(X_i - \mu)^2]}_{\text{I}} + \underbrace{\frac{2}{n-1} \sum_{i=1}^n E[(X_i - \mu)(\mu - \bar{X})]}_{\text{II}} + \underbrace{\frac{1}{n-1} \sum_{i=1}^n E[(\mu - \bar{X})^2]}_{\text{III}}$$

$$\begin{aligned} (a+b)^2 &= a^2 + 2ab + b^2 \\ a &= (X_i - \mu) \\ b &= (\mu - \bar{X}) \end{aligned}$$

$$\text{I} = \frac{1}{n-1} \sum_{i=1}^n E[(X_i - \mu)^2] = \frac{1}{n-1} \sum_{i=1}^n \text{var}(X_i) = \frac{1}{n-1} \sum_{i=1}^n \sigma^2 = \frac{n}{n-1} \sigma^2$$

$$\text{III} = \frac{1}{n-1} \sum_{i=1}^n E[(\mu - \bar{X})^2] = \frac{1}{n-1} \sum_{i=1}^n \frac{\sigma^2}{n} = \frac{\sigma^2}{n-1}$$

$$E[(\mu - \bar{X})^2] = E[(\bar{X} - \mu)^2] = E[(\bar{X} - E[\bar{X}])^2] = \text{var}(\bar{X}) = \frac{\sigma^2}{n}$$

We showed that $E[\bar{X}] = \mu$

$$\text{II} = \frac{2}{n-1} \sum_{i=1}^n E[(X_i - \mu)(\mu - \bar{X})]$$

$$= \frac{2}{n-1} E\left[\sum_{i=1}^n (X_i - \mu)(\mu - \bar{X})\right]$$

$$= \frac{2}{n-1} E[(\mu - \bar{X}) \sum_{i=1}^n (X_i - \mu)]$$

$$= \frac{2n}{n-1} E[(\mu - \bar{X}) \frac{\sum_{i=1}^n (X_i - \mu)}{n}] \quad \rightarrow \quad \frac{\sum_{i=1}^n (X_i - \mu)}{n} = \frac{\sum_{i=1}^n X_i}{n} - \frac{\sum_{i=1}^n \mu}{n} = \frac{\sum X_i}{n} - \frac{n\mu}{n}$$

$$= -\frac{2n}{n-1} E[(\bar{X} - \mu) \frac{\sum_{i=1}^n (X_i - \mu)}{n}]$$

$$= -\frac{2n}{n-1} E[(\bar{X} - \mu)^2] = -\frac{2n}{n-1} \text{var}(\bar{X}) = -\frac{2n}{n-1} \cdot \frac{\sigma^2}{n} = -\frac{2\sigma^2}{n-1}$$

$$E[S^2] = \text{I} + \text{II} + \text{III} = \frac{n}{n-1} \sigma^2 - \frac{2}{n-1} \sigma^2 + \frac{\sigma^2}{n-1} = \frac{n\sigma^2 - 2\sigma^2 + \sigma^2}{n-1} = \frac{(n-1)\sigma^2}{n-1} = \sigma^2$$

"sample variance"

"true" or population variance

② Hat problem: N people throw their hats into the center of a room. Then each person chooses one hat at random. X = # of ppl who get their own hat back. previously we showed that $E[X] = 1$. We did this by defining

① $X_i = \mathbb{1}\{\text{person } i \text{ got their own hat back}\}$

② $X = \sum_{i=1}^N X_i$ Don't need independence of X_i 's

③ $E[X] = \sum_{i=1}^N E[X_i] = \sum_{i=1}^N P(\text{person } i \text{ got their own hat back}) = \sum_{i=1}^N \frac{1}{N} = 1$

Now want to calculate $\text{var}(X)$

$$\text{var}(X) = \text{var}\left(\sum_{i=1}^N X_i\right) = \sum_{i=1}^N \text{var}(X_i) + 2 \sum_{i < j} \text{cov}(X_i, X_j) \quad (\text{Because } X_i \nparallel X_j)$$

$$\begin{aligned} \text{var}(X_i) &= \text{var}(\mathbb{1}\{\text{person } i \text{ got their own hat back}\}) \\ &= E[X_i^2] - E[X_i]^2 = \frac{1}{N} - \frac{1}{N^2} = \frac{N-1}{N^2} \end{aligned}$$

$$E[X_i] = P(\text{person } i \text{ got their own hat back}) = \frac{1}{N} \quad (\text{from previous lecture})$$

$$\begin{aligned} E[X_i^2] &= 1^2 P(\text{person } i \text{ got their own hat back}) \\ &\quad + 0^2 P(\text{person } i \text{ did not get their own hat back}) = \frac{1}{N} \end{aligned}$$

i ≠ j

$$\text{cov}(X_i, X_j) = E[X_i X_j] - \underbrace{E[X_i]}_{1/N} \underbrace{E[X_j]}_{1/N} = E[X_i X_j] - \frac{1}{N^2} = \frac{1}{N(N-1)} - \frac{1}{N^2} = \frac{1}{N^2(N-1)}$$

$$\begin{aligned} X_i X_j &= \mathbb{1}\{\text{person } i \text{ hat back}\} \mathbb{1}\{\text{person } j \text{ hat back}\} \\ &= \mathbb{1}\{\text{person } i \text{ hat back} \cap \text{person } j \text{ hat back}\} \end{aligned} \quad \mathbb{1}\{A \cap B\} = \mathbb{1}\{A\} \mathbb{1}\{B\}$$

$$\begin{aligned} E[X_i X_j] &= E[\mathbb{1}\{\text{person } i \text{ hat back} \cap \text{person } j \text{ hat back}\}] \\ &= P(\text{person } i \text{ hat back} \cap \text{person } j \text{ hat back}) \neq P(\text{person } i) P(\text{person } j) \\ &= P(\text{person } i \text{ hat back}) P(\text{person } j \text{ hat back} | \text{person } i \text{ hat back}) \\ &= \frac{1}{N} \cdot \frac{1}{N-1} \end{aligned}$$

Putting it all together...

$$\begin{aligned} \text{var}(X) &= \text{var}\left(\sum_{i=1}^N X_i\right) = \sum_{i=1}^N \text{var}(X_i) + 2 \sum_{i < j} \text{cov}(X_i, X_j) \\ &= \sum_{i=1}^N \frac{N-1}{N^2} + 2 \sum_{i < j} \frac{1}{N^2(N-1)} \\ &= \frac{N-1}{N} + \frac{N(N-1)}{N^2} \cdot \frac{1}{N-1} = \frac{N-1}{N} + \frac{1}{N} = \frac{N-1+1}{N} = 1 \end{aligned}$$

$$2 \sum_{i < j} = \sum_{i \neq j} = \underbrace{N^2}_{\text{All } i, j \text{ pairs}} - \underbrace{N}_{\text{pairs where } i=j} = N(N-1)$$

$$\sum_{i \neq j} \neq \sum_{i=1}^n \sum_{j=1}^{n-1} \quad (i, j) = (1, 1)$$

Conditional expectation

DEF: Suppose X and Y are r.v.s. The conditional expectation of X given $Y=y$ is denoted by $IE[X|Y=y]$ and is given by the following

If X and Y are discrete, Then

$$IE[X|Y=y] = \sum_x x P(X=x|Y=y) = \sum_x x \underline{P_{X|Y}(x|y)}$$

This is defined for all y such that $P(Y=y) > 0$

If X and Y are continuous Then

$$IE[X|Y=y] = \int x \underline{f_{X|Y}(x|y)} dx$$

$$P_{X|Y}(x,y) = \frac{P_{X,Y}(x,y)}{P_Y(y)}$$

This is defined for all y s.t. $f_Y(y) > 0$

$$f_{X|Y}(x,y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

Note: $IE[X|Y=y]$ returns a number for each value of y (s.t. The conditional dist is defined)

$$IE[X|Y=y] = h(y) \text{ (function of } y)$$

We say $\underline{IE[X|Y] = h(Y)}$ is the conditional mean of X given Y

This is a random variable

Properties of conditional expectation

① LOTUS:

$$IE[g(X)|Y=y] = \begin{cases} \sum_x g(x) P_{X|Y}(x|y) & \text{If } X, Y \text{ discrete} \\ \int g(x) f_{X|Y}(x|y) dx & \text{If } X, Y \text{ continuous} \end{cases}$$

② Linearity $IE[\sum_{i=1}^n X_i | Y=y] = \sum_{i=1}^n IE[X_i | Y=y]$

Law of total expectation + law of total variance

Recall: Law of total prob: $P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$

(Tower rule)

Law of total expectation: $IE[X] = IE[IE[X|Y]]$

Proof: Assume X, Y are discrete

$$\begin{aligned} IE[\underbrace{IE[X|Y]}_{\substack{\text{cond. exp. is a} \\ \text{function of } Y}}] &= IE[h(Y)] \quad \text{LOTUS} \\ &= \sum_y h(y) P_Y(y) \quad \text{we can write } h(y) = IE[X|Y=y] \\ &= \sum_y IE[X|Y=y] P_Y(y) \\ &\quad \text{By def of} \\ &\quad \text{conditional} \\ &\quad \text{exp.} \\ &= \sum_y \left(\sum_x x P_{X|Y}(x|y) \right) P_Y(y) \\ &= \sum_y \sum_x x \frac{P_{X,Y}(x,y)}{P_Y(y)} \cdot \cancel{P_Y(y)} \\ &= \sum_y \sum_x x P_{X,Y}(x,y) \end{aligned}$$

$$\text{(usually can interchange } \sum) = \sum_x \sum_y x P_{X,Y}(x,y)$$

$$= \sum_x x \sum_y P_{X,Y}(x,y)$$

$$\text{(def of marginal pmf)} = \sum_x x P_X(x) = E[X]$$

Law of total variance : If X and Y are r.v.s
 $\text{var}(Y) = E[\text{var}(Y|X)] + \text{var}(E[Y|X])$

Examples/problems

① Let N be a r.v. that takes values in $\{1, 2, 3, \dots\}$ and suppose $E[N] < \infty$.
 Let $X_1, X_2, \dots$ be a sequence of iid r.v.s that are independent of N
 and are such that $E[X_i] < \infty$. Let $Y = \sum_{i=1}^N X_i$

① What is $E[Y]$?

$$E[Y] = E\left[\sum_{i=1}^N X_i\right] \neq \sum_{i=1}^N E[X_i]$$

Because this random, cannot apply linearity right away

$$= E\left[E\left[\sum_{i=1}^N X_i \mid N=n\right]\right]$$

$$= E\left[E\left[\sum_{i=1}^n X_i \mid N=n\right]\right] \quad (\text{from conditionality})$$

$$= E\left[\sum_{i=1}^n E[X_i \mid N=n]\right] \quad (\text{linearity of cond. exp.})$$

$$= E\left[\sum_{i=1}^N E[X_i]\right] \quad (\text{independence of } N \text{ and } X_i)$$

$$= E[N E[X_i]] = E[X_i] E[N]$$

② What is $\text{var}(Y)$? We assume $\text{var}(N) < \infty$ and $\text{var}(X_i) < \infty$

$$\text{var}(Y) = \underbrace{E[\text{var}(Y|N)]}_{\text{I}} + \underbrace{\text{var}(E[Y|N])}_{\text{II}}$$

$$\text{II} : E[Y|N] = N E[X_i]$$

$$E[Y|N=n] = E\left[\sum_{i=1}^n X_i \mid N=n\right] = E\left[\sum_{i=1}^n X_i\right] = n E[X_i]$$

$$\text{var}(E[Y|N]) = \text{var}(N E[X_i]) = E[X_i]^2 \text{var}(N)$$

$$\text{I} : \text{var}(Y|N) = \text{var}\left(\sum_{i=1}^N X_i \mid N\right)$$

(independence)

$$= \sum_{i=1}^N \text{var}(X_i|N) = \sum_{i=1}^N \text{var}(X_i) = N \text{var}(X_i)$$

$$E[\text{var}(Y|N)] = E[N \text{var}(X_i)] = \text{var}(X_i) E[N]$$

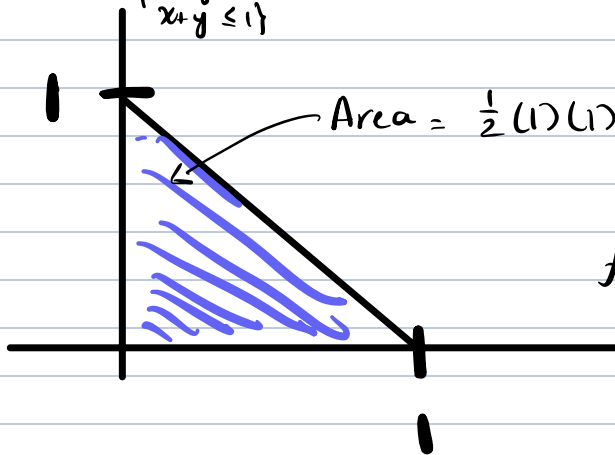
$$\text{var}(Y) = \text{I} + \text{II} = \text{var}(X_i) \mathbb{E}[N] + \mathbb{E}[X]^2 \text{var}(N)$$

(2) A random point (X, Y) is distributed uniformly over the triangle $\{(x, y) : x, y \geq 0, x+y \leq 1\}$

(a) Find $\mathbb{E}[Y]$ by finding the marginal distribution of Y

(b) Find $\mathbb{E}[Y]$ using tower property / law of total expectation

(a) Step 1: find the joint pdf
 $\int \int_C \{x, y \geq 0, x+y \leq 1\} dy dx = 1$



$$\text{Area} = \frac{1}{2}(1)(1) = \frac{1}{2} \Rightarrow \int \int_{\{x, y \geq 0, x+y \leq 1\}} dx dy = \frac{1}{2} \Rightarrow c = 2$$

$$f_{XY}(x, y) = 2 \mathbb{1}_{\{x, y \geq 0, x+y \leq 1\}}$$

Step 2: Find $f_Y(y) = \int 2 \mathbb{1}_{\{x, y \geq 0, x+y \leq 1\}} dx$
 $= \int_0^{1-y} dx = 2(1-y)$ defined for $0 < y < 1$ (picture)

$$\mathbb{E}[Y] = \int_0^1 2y(1-y) dy = \dots = \frac{1}{3}$$

(b) ① Show that $Y|X=x \sim \text{Unif}(0, 1-x)$

② Use ① to calculate $\mathbb{E}[Y|X]$

③ Use tower property $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y|X]]$